12

Magnetic Fields and E&M

Leave some space in your notes before this to complete 12.3.3

12.4

Amperes Law

12.4.1

Ampere's Law Defined

-Usage to derive the field strength of a long, current-carrying Wire

Ampere's Law

Ampere's Law relates the magnetic field created in space to the current enclosed by an imaginary closed path called an Amperian loop.

$$\boxed{\oint \vec{B} \circ \overrightarrow{d\ell} = \mu_0 I_{enc} + \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}}$$

- $\oint$ Closed line integral around the Amperian loop
- $\vec{B} \circ \overrightarrow{d\ell}$ Dot product of the magnetic field and a tiny element of the Amperian loop
- $\mu_0 I_{enc}$ The product of the permeability of free space (μ_0) and the current flowing through (enclosed by) the Amperian loop.
- $\mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$ "Maxwells extension" to Ampere's law. You are not expected to apply this in AP Physics C. You should understand that a changing electric field generates a magnetic field, which is what that term represents.

Ampere's Law

$$\oint \vec{B} \circ \overrightarrow{d\ell} = \mu_0 I_{enc}$$ On Equation Sheet

In AP Physics C, Amperes law may only be used in highly symmetric situations for which the line integral may be easily evaluated.

$$\oint B d\ell \cos\theta = \mu_0 I_{enc}$$ → 1. Simplify the dot product

$$B \cos 0 \oint d\ell = \mu_0 I_{enc}$$ → 2. Check for required symmetry

$$B \oint d\ell = \mu_0 I_{enc}$$ → 3. Evaluate the remaining integral. With everything else removed $\oint d\ell$ will simply become the length of the path (usually the perimeter)

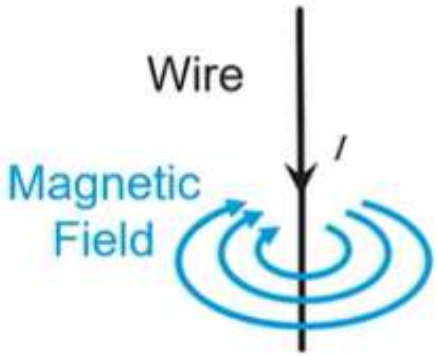
Let's look at an
EXAMPLE!

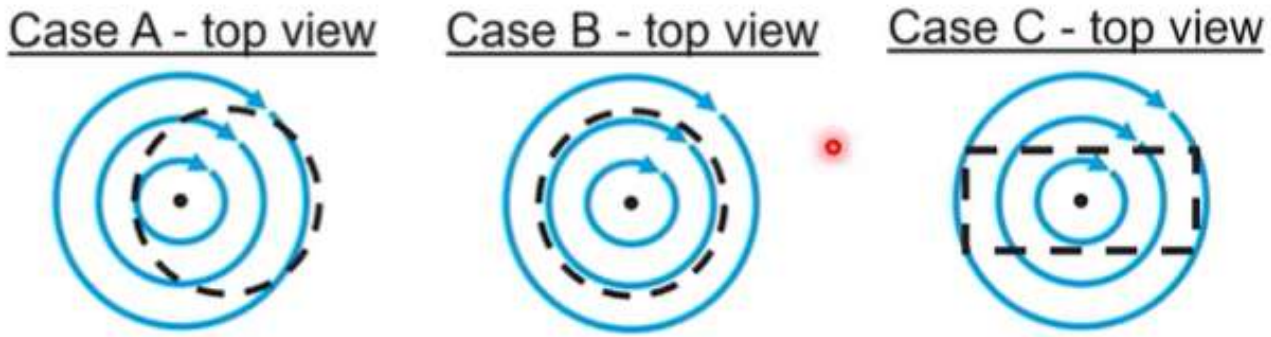
Ampere's Law

$$\oint \vec{B} \circ \overrightarrow{d\ell} = \mu_0 I_{enc}$$

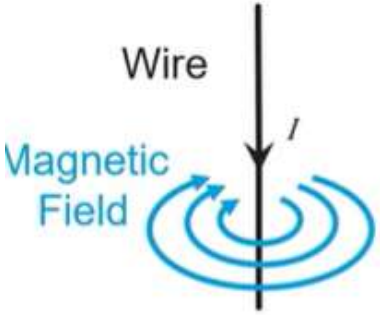
$$\oint B d\ell \cos\theta = \mu_0 I_{enc}$$

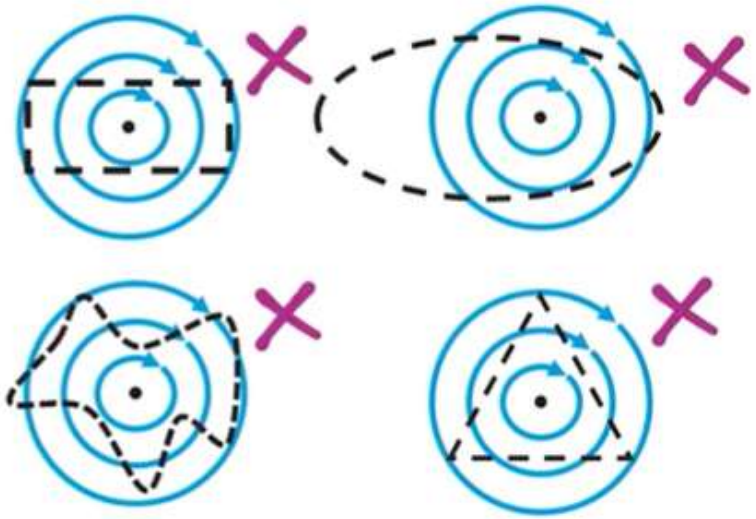
Symmetry is required for this level of Physics CEM!





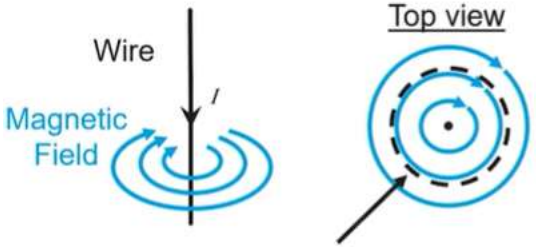
Top View of Various Amperian Loops





- YOU are in charge of creating an Amperian loop.
- There are an infinite number of possibilities (choose wisely)
- All loops satisfy Ampere's Law
- Only loops that meet the symmetry requirements should be used in this course

Derive Magnetic field of long straight current-carrying wire using Amperes Law



Amperian loop is created such that…

- Magnetic field is constant everywhere on loop (all points same distance from center)
- Magnetic field is parallel to loop at all points

$$\oint \vec{B} \circ \overrightarrow{d\ell} = \mu_0 I_{enc}$$

$$\oint B d\ell \cos\theta = \mu_0 I_{enc}$$

$$B \cos 0 \oint d\ell = \mu_0 I_{enc}$$

$$B \oint d\ell = \mu_0 I_{enc}$$

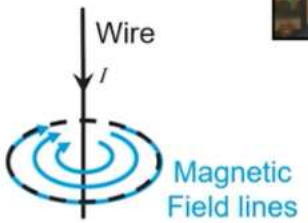
$$B(2\pi r) = \mu_0 I_{enc}$$

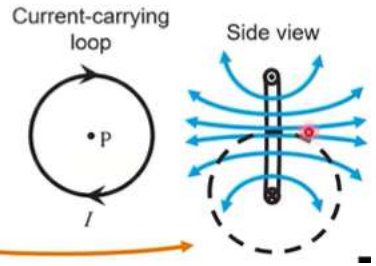
✓ $$\boxed{B = \frac{\mu_0 I_{enc}}{2\pi r}}$$

Matches result from video on Biot Savart Law

Takeaways

- Ampereian loop is chosen by you and must be a closed path.
- In general, line integrals are difficult to evaluate and mostly beyond the scope of this class.
- In certain situations, a carefully chosen Amperian loop will allow the line integral to be evaluated easily.
- A **good Amperian loop** has a magnetic field that is constant in magnitude and parallel to the Amperian loop or has the magnetic field perpendicular to the Amperian loop.
- You ***cannot* use Ampere's Law** to find the magnetic field at the center of a single current carrying loop. The magnetic field is neither constant nor parallel at every point on the loop.





12.4.2

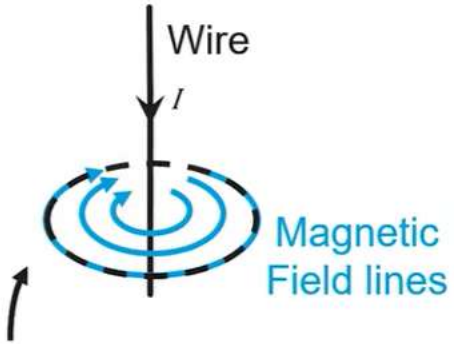
Using Ampere’s Law to graph the magnetic field of a Coaxial Wire

Ampere's Law

Ampere's Law relates the magnetic field created in space to the current enclosed by an imaginary closed path called an Amperian loop.

$$\oint \vec{B} \circ d\vec{\ell} = \mu_0 I_{enc}$$

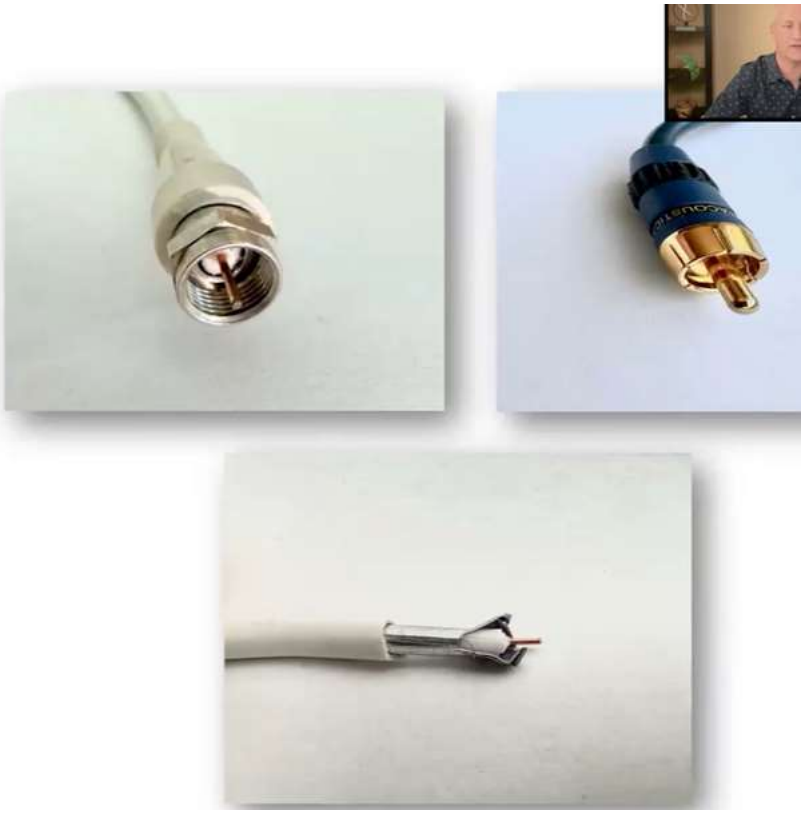
- $\oint$ Closed line integral around the Amperian loop
- $\vec{B} \circ \overrightarrow{d\ell}$ Dot product of the magnetic field and a tiny element of the Amperian loop $\overrightarrow{d\ell}$
- $\mu_0 I_{enc}$ The product of the permeability of free space (μ_0) and the current flowing through (enclosed by) the Amperian loop.



- A **good Amperian loop** has…
  - a magnetic field that is constant in magnitude and parallel to the Amperian loop
  - If the above condition is not met, then the magnetic field must be perpendicular to the loop so that the dot product will become zero.

What is a Coaxial Wire?

- A type of electrical cable.
- Features an inner conducting core, separated from an outer concentric conducting sheath.
- Used for signal transmission such as cable TV, audio, internet, radio antenna connections, musical instruments.
- Patented in Britan by Oliver Heaviside in 1880 (also known for developing the field of vector analysis and rewriting Maxwell's equations as we know them today).

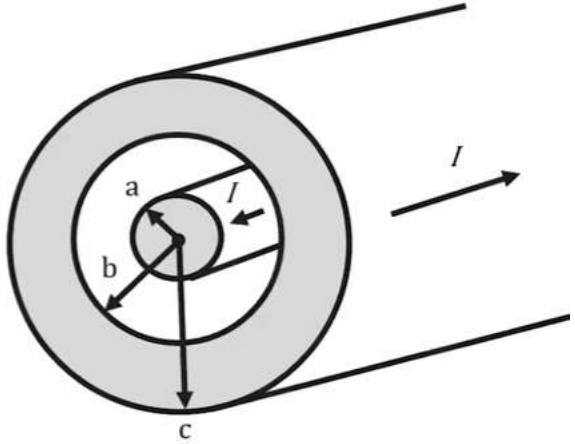


Magnetic Field of a coaxial wire

Sketch a graph of the magnetic field of a current carrying coaxial wire as a function of radius.

$$\oint \vec{B} \circ \overrightarrow{d\ell} = \mu_0 I_{enc}$$

- Use Ampere's Law to determine the magnetic field as a function of radius in each of 4 distinct regions.
- Inner core $r < a$ requires using current density of core to find I_{enc}
- Gap between wires $\mathrm{a} < r < b$ enclosed current is $+I$
- Inside outer sheath $\mathrm{b} < r < c$ requires using current density of outer sheath to find I_{enc}
- Outside everything $\mathrm{r} > \mathrm{c}$ $\quad I_{enc} = 0$



Region 1 – inner wire

- Inner core $r < a$ requires using current density of core to find I_{enc}
- Define direction of current flow. Out of page is positive, into page is negative

$$\oint \vec{B} \circ \overrightarrow{d\ell} = \mu_0 I_{enc}$$

$$B \oint d\ell = \mu_0 J A_{enc}$$

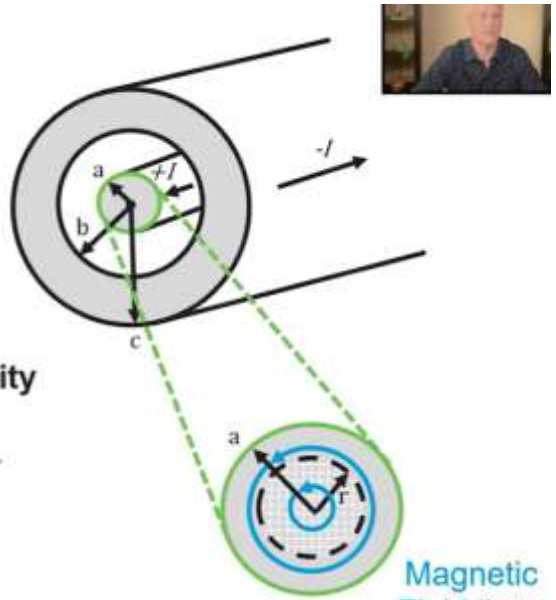
$$B(2\pi r) = \mu_0 \frac{+I}{\pi a^2}(\pi r^2)$$

$$\boxed{B = \frac{\mu_0 I r}{2\pi a^2}}$$

B increases linearly in core

Current Density

$$J = \frac{I}{A} = \frac{+I}{\pi a^2}$$



Region 2 – gap between wires

- $a < r < b$
- enclosed current is $+I$

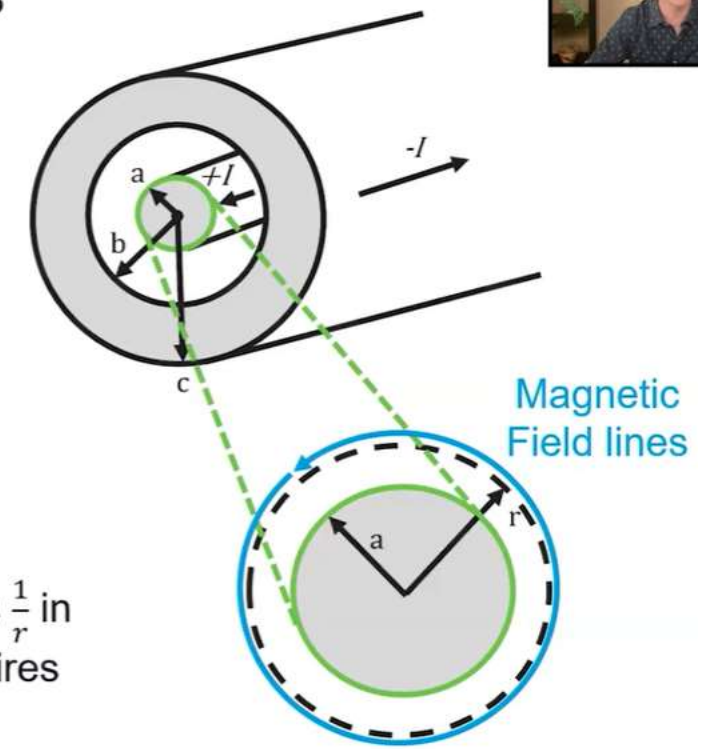
$$\oint \vec{B} \circ \overrightarrow{d\ell} = \mu_0 I_{enc}$$

$$B \oint d\ell = \mu_0 I$$

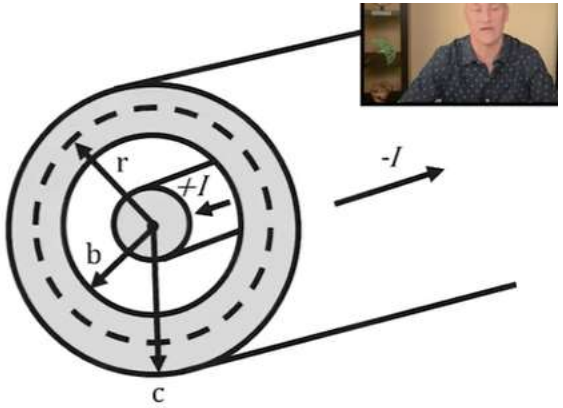
$$B(2\pi r) = \mu_0 I$$

$$\boxed{B = \frac{\mu_0 I}{2\pi r}}$$

B decreases as $\frac{1}{r}$ in gap between wires



Region 3 – inside outer sheath



- Inside outer sheath $b < r < c$ requires using current density of outer sheath to find I_{enc}
- Negative current in outer sheath

Current enclosed

$$I_{enc} = I - I_{enc-outer}$$

$$= I - J_{outer} \cdot A_{enc-outer}$$

$$= I - \frac{I}{\pi c^2 - \pi b^2} \cdot (\pi r^2 - \pi b^2)$$

$$= I - I\frac{r^2 - b^2}{c^2 - b^2}$$

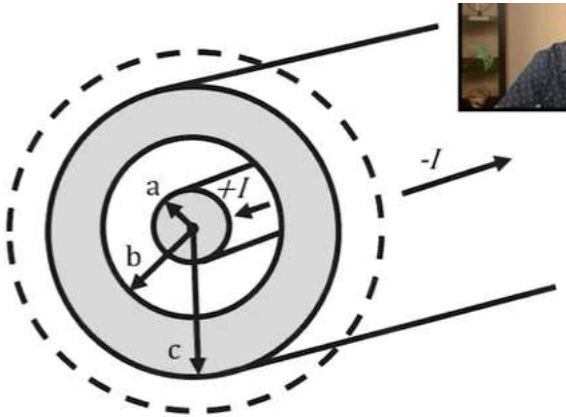
$$\oint \vec{B} \circ \overrightarrow{d\ell} = \mu_0 I_{enc}$$

$$B(2\pi r) = \mu_0 \left(I - I\frac{r^2 - b^2}{c^2 - b^2} \right)$$

$$\boxed{B = \frac{\mu_0 I}{2\pi r}\left(1 - \frac{r^2 - b^2}{c^2 - b^2}\right)}$$

B decreases as a concave up curve

Region 4 – outside of everything



- Outside coaxial wire $r > c$

$$\oint \vec{B} \circ \overrightarrow{d\ell} = \mu_0 I_{enc}$$

$$B \oint d\ell = \mu_0 (I - I)$$

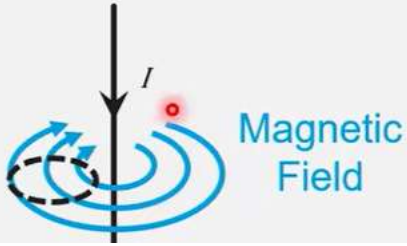
$$B \cdot 2\pi r = 0$$

$$\boxed{B = 0}$$

No magnetic field outside due to symmetry and nothing else nearby to create a magnetic field

If $I_{enc} = 0$ it will not always follow that $B = 0$.

This Amperian loop has $I_{enc} = 0$, yet there is still a magnetic field. However, it is still true that $\oint \vec{B} \circ \overrightarrow{d\ell} = 0$



Sketch the graph of magnetic field vs. radius

When $r < a$, B increases linearly

$$B = \frac{\mu_0 I r}{2\pi a^2}$$

When $\mathrm{a} < r < b$, B decreases as $\frac{1}{r}$

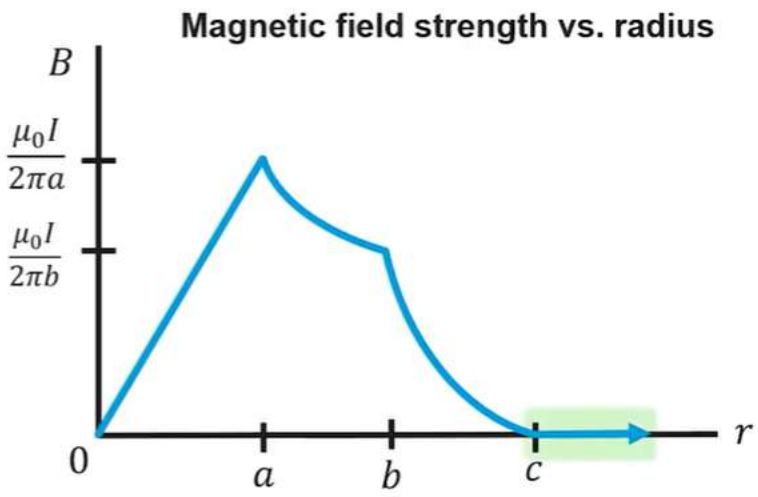
$$B = \frac{\mu_0 I}{2\pi r}$$

When $\mathrm{b} < r < c$, B decreases concave up

$$B = \frac{\mu_0 I}{2\pi r}\left(1 - \frac{r^2 - b^2}{c^2 - b^2}\right)$$

When $r > c$, There is no magnetic field

$$B = 0$$



Takeaways

- Using Ampere's Law to find magnetic field strength depends on symmetry.

- To simplify $\oint \vec{B} \circ d\vec{\ell}$ check the following conditions.
  - Magnetic field is constant over the integration path (unless $\theta = 90°$)
  - Magnetic field is at a constant angle to the integration path (usually parallel 0°).

- When working inside a wire, the enclosed current can be found using current density and enclosed area.

$$I_{enc} = J \cdot A_{enc}$$

- When sketching a graph, you do not need to include units, scale your axis or plot data points. Your graph should show the correct relationships (be attentive to the shape). Labels of intercepts, maxima, minima and asymptotes may be required.

12.4.3

Using Ampere’s Law to calculate the number of turns required in a Solenoid

What is a Solenoid?

A solenoid is an electrical device formed out of a tightly wound helical coil of wire.

- The length of the solenoid is typically much larger than its diameter.
- Each individual loop contributes to the magnetic field inside the coils.
- When current flows, a solenoid becomes an electromagnet.

Solenoids are commonly used in a variety of applications including automotive gearbox systems, electrical locks, switches and valves, doorbells, dialysis and MRI machines.

Later in this class you will study how solenoids act in DC circuits. In this context a solenoid is called an inductor.

No Current

Demonstration of magnetic field of solenoid

Current Applied

Demonstration of magnetic field of solenoid

- No magnetic field outside of solenoid.
- Magnetic field inside is parallel to axis of solenoid when current is flowing.

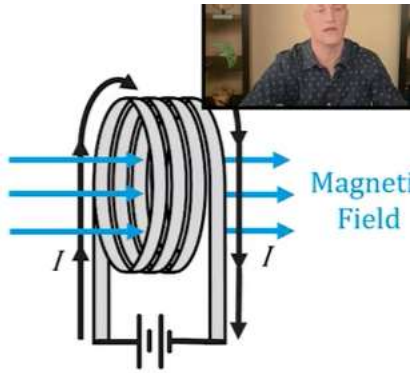
Assumptions made when modeling solenoids

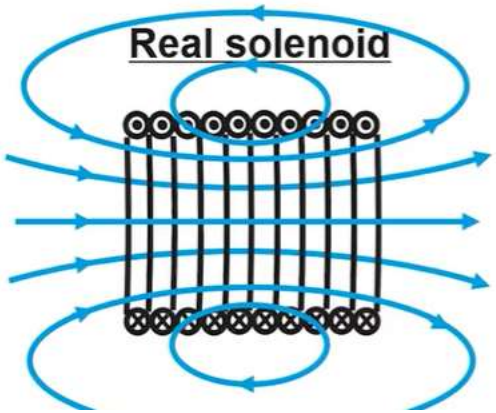
Ideal Solenoid	Real Solenoid
• Inside the solenoid, the magnetic field is uniform at all locations and parallel to the axis of the solenoid. • The magnetic field outside the solenoid is zero. • The length of the solenoid is much larger than its diameter. A truly ideal solenoid would have infinite length. • Coils are wound very tightly without any gaps. • No resistance	• The magnetic field is strongest at the center and weaker at the ends. • There is a weak magnetic field outside the solenoid. • Experiences “edge effects” where the strength and direction of the magnetic field will vary towards the ends. • The length of the solenoid is often larger than its diameter but not always. • Has some resistance

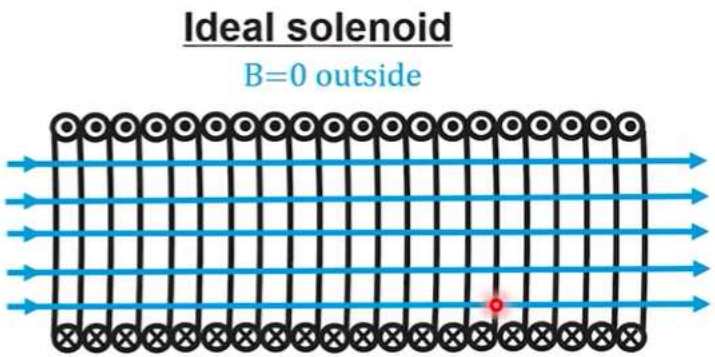
Magnetic field of a solenoid

To find the direction of the magnetic field…

- Use your right hand
- Curl your fingers in the direction current flows through the loops
- Your thumb will point in the direction of the magnetic field inside the solenoid





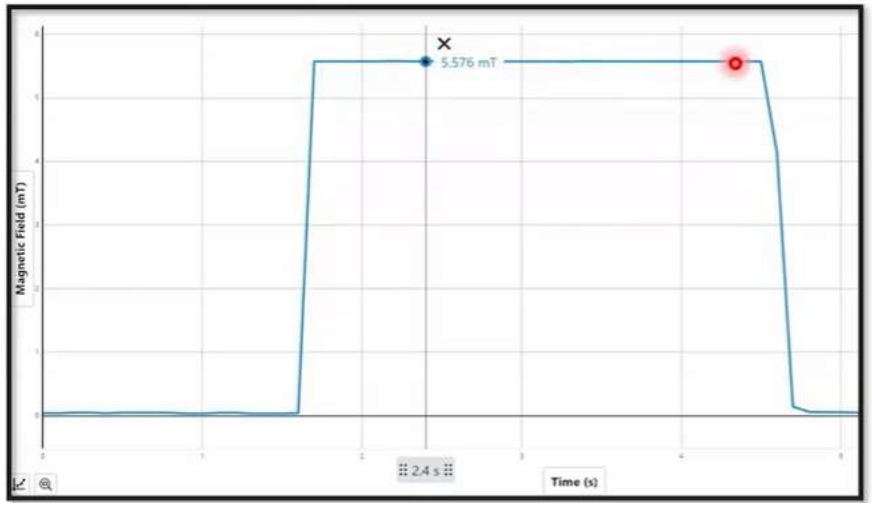


How many turns are there?

Current = 2.33A

Length of Solenoid = 11.5cm = 0.115m

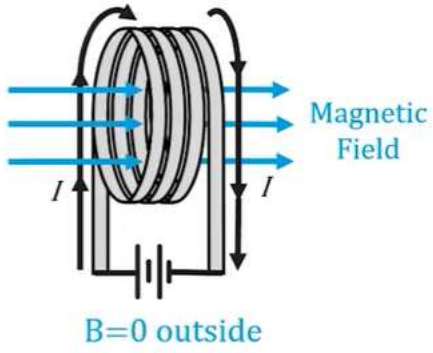
Magnetic Field Strength = 5.78mT (at end of solenoid)



Let's make a plan to find the number of turns

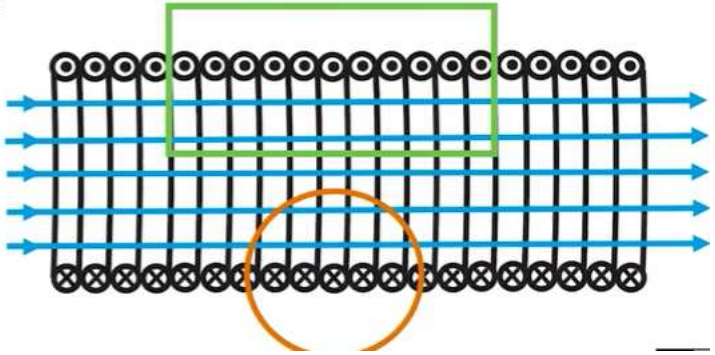
How should we start?

1. Choose an appropriate Amperian loop.
   a) B field must be constant in magnitude and parallel or perpendicular to the loop at all points.
2. Determine the amount of current enclosed by the loop.
3. Apply Ampere's Law.
4. Simply and solve for the number of turns.

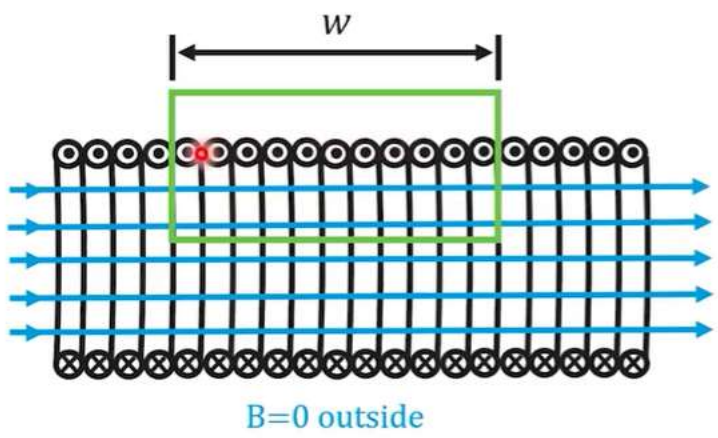


✓ The best Amperian loop for this problem is the one in green.

✗ The Amperian loop in orange is a poor choice because the angle the magnetic field makes with the loop is changing.



Let's find the current enclosed



Each loop of wire has the same current flowing through it.

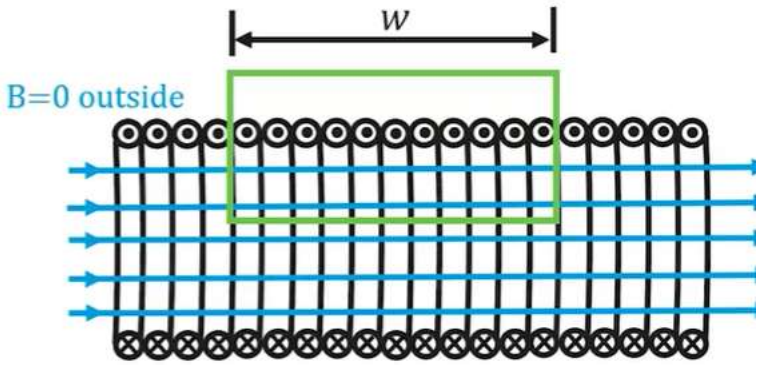
$$I_{enc} = I \cdot n \cdot w$$

n = number of turns per unit length of the solenoid.

w = the width of the Amperian loop (rectangle).

Let's apply Ampere's Law

$$\oint \vec{B} \circ \overrightarrow{d\ell} = \mu_0 I_{enc}$$



The line integral will get split up into 4 distinct parts, one for each of the 4 sides of the Amperian loop.

$$\underbrace{\int \vec{B} \circ d\vec{\ell}}_{\text{Left}} + \underbrace{\int \vec{B} \circ d\vec{\ell}}_{\text{Right}} + \underbrace{\int \vec{B} \circ d\vec{\ell}}_{\text{Top}} + \underbrace{\int \vec{B} \circ d\vec{\ell}}_{\text{Bottom}} \longrightarrow BCos(0°) \int d\vec{\ell} = \mu_0 I_{enc}$$

$$B \cdot w = \mu_0 (I \cdot n \cdot w)$$

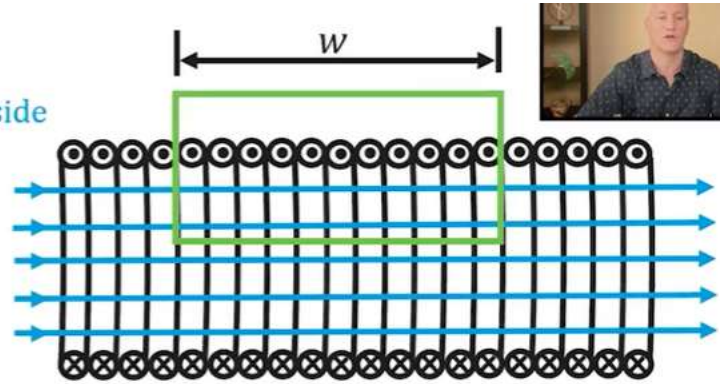
$$B_{sol} = \mu_0 n I$$

The left and right sides have a 90° between $\vec{B}$&$d\vec{\ell}$ so the dot product is 0.

The top of the loop has no magnetic field and if B=0, the dot product is 0.

At the bottom of the loop the magnetic field is constant and parallel to loop.

Let's find the number of turns



$$B_{sol} = \mu_0 n I$$

Remember "n" is the number of turns per unit length

$$n = \frac{Total\ number\ of\ turns}{Total\ length\ of\ solenoid} = \frac{N}{L}$$

$$B_{sol} = \mu_0 \frac{N}{L} I$$

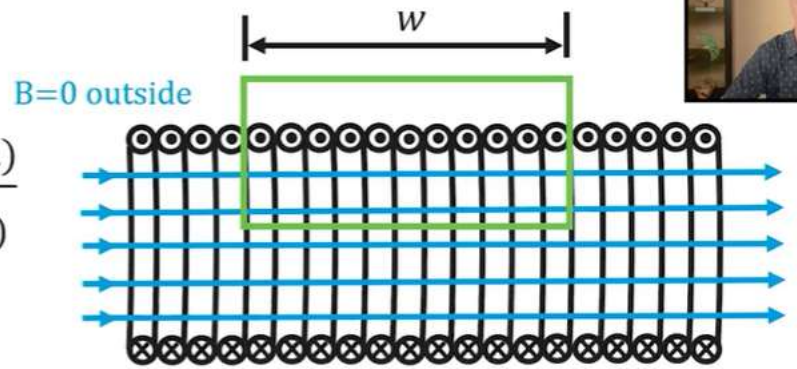
$$\frac{B_{sol} L}{\mu_0 I} = N$$

Plug in the known quantities

$$N = \frac{B_{sol} L}{\mu_0 I} = \frac{(5.78 \cdot 10^{-3} T)(0.115 m)}{(4\pi \cdot 10^{-7} \frac{Tm}{A})(2.33 A)}$$

N =227
Total number of turns in solenoid

Error Analysis



$$N = \frac{B_{sol}L}{\mu_0 I} = \frac{(5.78 \cdot 10^{-3}T)(0.115m)}{(4\pi \cdot 10^{-7}\frac{Tm}{A})(2.33A)}$$

$N = 227$ turns

The actual number of turns is 250.

Can you name a physical reason that would lead to the calculated number of turns being less than the actual number?

1. The length measured was shorter than the actual length.
2. The current shown on the meter was larger than the actual current.
3. The magnetic field is weaker just outside the end of the solenoid than it would be in the center.

- A solenoid is made from a long wire coiled many times to form a tube.
- Solenoids are electromagnets and have many industrial applications.
- Ideal solenoids have no resistance, a constant magnetic field inside and no field outside.
- Amperian loops can be any shape, not just circles.
- Circles and rectangles are the Amperian loops most common in AP Physics C.
- There is a difference between the total number of turns of a solenoid and the number of turns per unit length.

$$B_{sol} = \mu_0 n I$$
$$n = turns\ per\ unit\ length$$